SRINIVASAN MANI

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PROFILE SUMMARY

Geospatial Data Engineer with 6+ years of experience working with large-scale satellite and sensor datasets, focusing on time-series analysis, spatial data processing, and scalable data pipelines. Strong background in Python-based data engineering, geospatial systems, and scientific computing, with hands-on experience in GNSS data, electron density modeling, and real-time data workflows. Particularly interested in anomaly detection, multi-sensor data fusion, and developing intelligent systems for situational awareness in complex environments.

PROFESSIONAL EXPERIENCE

DATA SCIENCE INTERN - GERMAN AEROSPACE CENTER, NEUSTRELITZ ***MAR 2025 – SEP 2025***

- Built the NEDM Interactive Ionospheric Visualisation Application using Marimo, enabling researchers to dynamically explore electron density distributions in real time. Optimized the backend with parallel processing to handle high-volume API requests across large geographic areas.
- Overhauled the Scintillation ETL pipeline by migrating from Pandas to Polars. Engineered logic to handle 64-column GNSS datasets including S4 indices, amplitude scintillation and satellite geometry parameters, then integrated the output into a containerized PostGIS database.
- Automated GeoServer OGC geospatial service publishing (WFS and Features API) using the REST API, replacing manual GUI workflows with programmatic workspace, datastore, and feature-type creation. Deployed within a containerized environment to ensure portability and consistency across DLR.
- Designed a PostgreSQL/PostGIS schema for Total Electron Content (TEC) maps and scintillation data, using JSONB for Ionospheric Pierce Points and grid models. Benchmarked ingestion strategies and identified multiprocessing as the most efficient approach for high-speed data loading.

SENIOR GIS ANALYST - TRIMBLE, INDIA ***JAN 2019 – FEB 2023***

- Led analytics and data integration efforts for large datasets supporting benchmarking and performance monitoring in nationwide logistics operations networks using spatial modeling, geofencing and exposure analysis which reducing delivery times by 50%.
- Deployed the COVID-19 Driver Information System, providing real-time store updates to 500+ drivers and ensuring safe logistics operations during the pandemic.
- Mentored a team of GIS analysts, maintaining data accuracy above 99% across nationwide spatial datasets.

GIS ENGINEER - TRIMBLE, INDIA ***APR 2017 – DEC 2019***

- Analyzed, validated, and maintained truck routing networks for U.S and Mexico operations, incorporating road restrictions, bridge clearances, speed limits and HazMat routes to ensure regulatory requirements.
- Enhanced routing and navigation systems across 25,000+ miles of transportation networks, increasing reliability and reducing routing errors.
- Produced maps, spatial reports, and dashboards to support operational and strategic planning.

INTERN - GYOKAP SURVEYING TRAINING INSTITUTE, INDIA ***DEC 23-31, 2019***

- Conducted field surveys using DGPS and UAV platforms for environmental and topographic data.
- Assisted with equipment setup, positioning, and basic calibration during field campaigns.
- Supported terrain modeling and quality checks of collected field data.

INTERN - INSTITUTE OF REMOTE SENSING (IRS), INDIA ***JUN 17 – JUL 01, 2014***

- Participated in land-use and land-cover field surveys integrating Total Station, DGPS, and satellite imagery.
- Supported ground-truth data collection and validation for environmental mapping projects.

INTERN - INDIAN METEOROLOGICAL DEPARTMENT (IMD), INDIA ***JUN 30 – JUL 04, 2014***

- Assisted in monitoring meteorological parameters using rain gauges and Doppler radar observations.
- Supported data logging and basic analysis for short-term environmental monitoring and forecasting.

EDUCATION

MASTER OF ENGINEERING IN GEOMATICS ***MAR 2023 – MAR 2026***

Hochschule Neubrandenburg, University of Applied Sciences, Germany

Focus: Remote Sensing, EO Data Analysis, Environmental Modeling
Thesis: Land use/land cover modeling and forecasting using EO data
Tools: Google Earth Engine, Sentinel-2, QGIS

BACHELOR OF ENGINEERING IN GEO-INFORMATICS ***SEP 2011 – APR 2015***

Regional Centre of Anna University, Tirunelveli, India

CONFERENCES

INTERGEO (2023), MESSE BERLIN, GERMANY

Attended a leading international conference on geodesy, GIS, UAV mapping, and 3D modeling technologies for smart cities and digital twins.

TECHNICAL SKILLS

Programming	Python, JavaScript, HTML, CSS (Basic knowledge of R)
Data Engineering & Management	PostgreSQL/PostGIS, ETL pipelines, spatial data modeling, WKT queries, handling large-scale datasets
Distributed Systems & APIs	REST APIs, GeoServer, OGC/WFS services, STAC concepts
Data Analysis & Simulation	Spatiotemporal analysis, statistical modeling, simulation-oriented workflows, model validation
Transport & Geospatial Systems	Transport network data, routing systems, GNSS datasets (S4, TEC, IPP), vector/raster processing
Geospatial Tools & Platforms	QGIS, ArcGIS Pro, Google Earth Engine, GeoServer, PostGIS
Scientific & Environmental Data	NetCDF (basic), Earth observation data processing, spatial data integration
DevOps & Workflows	Docker, Git, CI/CD, Linux/CLI, parallel processing, Marimo

ADDITIONAL INFORMATION

Languages	English (C1), German (A2), Tamil (Native)
Soft Skills	Problem-solving, team collaboration, research communication

REFERENCES

Prof. Dr. Frank Günther	Professor, Photogrammetry, Remote Sensing, GIS, Cartography Hochschule Neubrandenburg, University of Applied Sciences Email: `guenther@hs-nb.de`
Martin Kriegel, M.Eng.	Group Leader, Pre-operational Services German Aerospace Center (DLR), Neustrelitz Email: `Martin.Kriegel@dlr.de`